

Engineering conformance specification

v0.1/r0.3

Scientific owner: Grant Wilson

Status: current review candidate; not owner approved, frozen or activated.

Date: 2026-09-08. Numerical methods/routes: `unavailable_pending_separate_owner_approval`.

Document identity

SCI-FRUIT-ENGINEERING-CONFORMANCE v0.1/r0.3

Frozen Stage A packet

SCI-FRUIT-SCIENTIFIC-CORE-STAGE-A-R0.4-AMENDED

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Targeted revision directive

SCI-FRUIT-OD-STAGE-B-R0.3-FINAL-NARROW-REVISION-2026-09-08

Assigned record identifier for exact supplied directive bytes.

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Stage A approval / launch

SCI-FRUIT-OD-STAGE-A-R0.4-AMENDED-APPROVAL-2026-09-07

SHA-256: 36fb05922951e77b4540c95192f639685b303a00d2831b772d83fb3ce78adb94

Stage A freeze

SCI-FRUIT Stage A r0.4 amended owner freeze, 2026-09-07

SHA-256: 754e81b3940f77f3bca067ba00a94f87707746d3c0bc9c6089487602de4c3aa8

Preserved r0.2 review-candidate delivery

SCI-FRUIT-v0.1-stage-b-r0.2-owner-review.tar.gz

SHA-256: 6d552e49510c92fff65d9c835f8f31cd21f8ef50bed4f1a24ff355cfce7523a4

Sole current review-candidate normative core

SCI-FRUIT-NORMATIVE-CORE v0.1/r0.3

Canonical source-inventory SHA-256: e6c3a536246e75739a50741de7862d664cd1ffd52ac877d5ef6b9ead45558c17

This document's source identity

SCI-FRUIT-ECS-DOCUMENT-SOURCES v0.1/r0.3

Document source-inventory SHA-256: c6d4c0dd8725159f45398e78f9c5ffaf585152b4368b5fa52469d4a268ab45c0

Build record

SCI-FRUIT-DOC-BUILD-R0.3-2026-09-08

Hashes bind exact inventory/control bytes, not a PDF containing its own digest. The source-inventory scopes and reproducible build recipe are delivered with this document. Document verification supplies no numerical conformance, qualification, validation, readiness or production authority.

1 The exact conformance candidate

This is a prospective engineering evidence procedure. Its sole scientific authority is SCI-FRUIT-NORMATIVE-CORE v0.1/r0.3 with the exact digest on this cover. It adds no numerical method or scientific obligation. Evidence is assessed per requirement for an immutable candidate, never for an unspecified implementation name or generic algorithm family.

Define the candidate tuple $\mathcal{C}$ with every component explicitly bound:

1. implementation/source identity, revision and exact source-tree digest;
2. build recipe/toolchain/artifact identity and runtime/environment identity;
3. normative-core identity and exact canonical source-inventory digest;
4. method-specific FRUIT-FEEDBACK-METHOD identity, complete record digest and owner-decision binding;
5. exact producer parent route, grouping, scientific identity and generation;
6. separate candidate/accepted/applied model generations and transformation bindings;
7. resolved ordered operation graph and all consequential state generations;
8. absolute iteration, lineage, branch, application and attempt identities;
9. output product role and immutable product generation;
10. evidence generation, with fixture/oracle/comparison and reviewer bindings.

Each identity must resolve to immutable bytes or an exact content-addressed scientific record. A candidate digest binds the complete tuple, its serialization convention and referenced objects; changing a component creates a new candidate. Do not use a filename, process exit or common array shape as an identity proof.

Before numerical evidence, check the method's exact separate owner approval and required upstream numerical authority. They are absent in this Stage B packet. Every numerical result here is **blocked**, explicitly dependency-blocked by the absent approved method/route, not **pass**. Document integrity is reported separately and cannot fill a candidate's numerical fields.

The blank candidate/result template accompanies this specification. Null fields are unresolved placeholders and are not a partially admitted candidate. A future assessment may instantiate them only within separately authorized work.

Normative source: review-candidate core, REQ-001–003, REQ-007, REQ-010, REQ-023.

2 Requirement evidence, results and comparisons

One evidence record binds requirement ID; complete candidate identity; exact observable; immutable fixture identity/digest; independent oracle or expected-value construction; preregistered comparison; result; artifact identities/digests; reviewer/date; and deviation/failure disposition. The record also retains blocking dependency, applicability premise, affected claim/scope and any superseded evidence parent. A requirement summary or a finite output is not evidence that the requirement holds.

Use exactly five result values, independently of evidence-artifact realization:

pass

The declared applicable requirement is supported for the exact candidate by complete admissible evidence under its preregistered comparison.

fail

Admissible evidence contradicts the applicable requirement or the candidate violates its mandatory record/gate contract; retain the exact discrepancy and scope.

blocked

A named required authority/input/fixture/oracle/comparison dependency prevents assessment; retain exact identity/cause and never count it as pass.

not_assessed

The applicable assessment has not been performed or evidence remains unassessed, without substituting for a known blocked dependency.

not_applicable

The core and exact method permit the absent role, with an explicit applicability premise and reviewer disposition. Missing input/approval is not inapplicability.

An assessment cannot claim overall conformance while applicable requirements are failed, blocked or unassessed. No required role can be waived by an engineering summary. Keep the coverage matrix and every failed/unavailable comparison, including an earlier attempt superseded by a corrected candidate. Evidence-artifact realization is separately **realized**, **incomplete**, **failed** or **not_produced**, with generation/cause/last reached stage. A realized accounting record may report blocked conformance; a missing evidence artifact does not become a scientific availability value or automatic pass.

Preregister the observable quantity, units, axes/domain/support, reference/gauge, comparison population, deterministic or stochastic target, oracle construction, finite/non-finite policy, numerical profile, comparison statistic and any tolerance, scale/denominator, sample-size or decision rule required by the approved method. This ECS selects none of those numerical values. If unavailable, block that numerical comparison by its exact missing dependency.

Oracle independence means the expected construction does not merely call the candidate implementation's same path. A symbolic deduction, separately derived operator calculation or immutable independently constructed record may serve only within its exact premises. Declare shared assumptions and dependencies. Use negative controls for missing/incompatible inputs and scope changes; passing one limiting case never proves general numerical fidelity or coverage.

Normative source: review-candidate core, REQ-004-006, REQ-021, REQ-023-024.

3 Parent, model transformation and operation evidence

EV-001: parent/type/method binding. Resolve every candidate tuple link and compare the method record, owner decision, producer route/generation, grouping and original measured-parent ancestry against an independently built identity ledger. Check input-rule conversions and all type fields. Negative fixtures vary one identity/domain/authority at a time. Missing numerical authority prevents execution-based conformance; do not substitute the family record. Retain the exact source/build candidate used by each observation.

EV-002: accepted-to-applied transformation. Bind candidate construction, selection, acceptance and application as distinct records. Observe the actual operands supplied to removal and rejoin. Independently construct the approved accepted-to-applied transformation, with normalization/sign/support/unit changes and the approved dependence rules. Compare objects, domains, supports, generations and separately addressable transformation uncertainty. Include a case where accepted and applied differ. Do not silently copy accepted covariance or error into applied axes. Numerical construction awaits the approved method.

EV-003: order/bypass/counts. Compare the complete ordered realized operation trace against the independently declared method graph. Check each path's removal/rejoin locations, every bypassed operation, application count, owner and state generation. Use fixtures with omitted, repeated or reordered operations as discriminators; these are prospective fixture designs, not executed instrumented reductions. An unchanged output alone does not excuse wrong scientific lineage or a duplicate response application.

EV-004: additive compatibility. Independently enumerate removal and rejoin operand-pair quantity/unit/scale/frame/grid/domain/support/reference/ gauge/null-space/calibration/response bindings. Check each exact declared conversion before arithmetic. Include equal-shape but incompatible-gauge or quantity records. The expected construction is the typed compatibility relation, not automatic array broadcasting or an inferred offset. Missing facts require typed unavailability for the affected composition.

For each procedure, emit individual requirement records using the evidence index. Attach immutable actual/expected records and discrepancy locations. Do not combine a successful identity check with an unassessed numerical result under one pass flag.

Normative source: review-candidate core, REQ-001–007, REQ-010–011, REQ-015.

4 Support, failure, state and limiting-case evidence

EV-005: support/origin. Compare all eight roles against independent path-domain and ancestry constructions, including three separate model-stage supports and learning influence. The fixture family distinguishes compatible both-path, permitted residual-only, permitted model-only and unavailable occurrences. Verify model/prior mode origin independently of those classes. Include an unavailable upstream absolute mode, so finite rejoin values cannot be mislabeled acquired exposure or recovered measurement. Preserve exact producer admission boundaries and absent JINC companion products. For the admitted PTC use, bind application/output support, all required-input and normal-matrix finiteness, and required rank under frozen tolerance. A perturbed object outside that exact domain yields affected unavailability or a separately typed transition; it does not silently become RF-03.

EV-006: fail-closed inputs and limiting equations. Supply prospective missing/incompatible parent, applied operand, contribution, support or additive reference fixtures. Expected records name the affected claim/domain/cause and last reached lifecycle stage; they do not invent zero, identity, bypass or fill. For numerically approved fixtures, independently substitute the precise zero-preserving, identity or complete-linear premises into the core equations (PRED-001–004). Compare each required path contribution as well as final output. Offsets and missing premises are negative controls. These deductions define expected constructions; no numerical fixture is instantiated here. For PRED-006, substitute the admitted modal-coordinate expression into the PTC output; compare fixed-map multiplication with frozen-component subtraction. Retain the fixed affine centering term and compare derivatives HP versus H. An internal-coordinate change with unchanged defining state is a fixed-map case; a prescribed rerun of defining-state resolution is a separate procedure case.

EV-007: rule/state/generation identity. Compare identical-rule changed realizations and changed-rule scenarios against an independent identity graph. Check all seven operation/state categories at their exact conditioning scopes, including fixed structure with recomputed operator-defining coefficients. Distinguish fixed-map internal-coordinate values from defining-state resolution; changed internal values alone create no new operator generation. Factor mixed operations rather than hiding their dependencies. Each distinct application/attempt has its bound states, lineage and parentage. A storage-only representation change must resolve to the same declared scientific state queries.

For all three procedures, require observables sufficient to assess the named scientific role. A single final array cannot expose missing support, hidden learning influence or a silently changed rule. If the candidate cannot expose or exactly reconstruct required evidence, report the missing observation as **blocked** with cause, or **fail** where an established mandatory record obligation is violated. Do not infer an underlying implementation behavior from the missing record alone.

Normative source: review-candidate core, REQ-005–011, REQ-018, REQ-021, REQ-024.

5 Response evidence with fixed maps and defining-state reruns

EV-010. Declare RF role, source/domain/units/basis, baseline, codomain, convention and support/reference/gauge/null-space comparison. Build independent records for operator-defining state, internal application coordinates and any frozen removed component. Bind their exact source/state identities.

For RF-01 perturb iteration input and freeze the complete induced maps, operator-defining state, applied model and other consequential fixed facts. For RF-02 perturb only the applied model with measured input fixed and the same map/state discipline. Observe internal-coordinate evaluations at both operands; changed values are permitted when their defining map stays fixed. No trace may silently freeze a required internal coordinate or rerun a frozen defining-state rule. Compare with the compatible fixed-map difference or its differentiable chain rule; complete linearity alone permits the A/D reduction.

For the exact admitted PTC specialization, independently substitute the modal coordinate into the output equation and bind $P_{g,t,\Theta}^{\text{PTC}}$. Compare δxP or HP under the same frozen defining facts. Preserve $-\lambda_g P$ in the uncentered affine output. A frozen-component family instead expects identity derivative H . Check all required input/normal-matrix finiteness, numerical rank under frozen tolerance and application/output support. No numeric fixture, tolerance or finite-difference step is chosen here.

For RF-03 trace every prescribed operation that can change defining maps or consequential state: model construction/acceptance, loading, metric, support/mask, centering/reference, rank, grouping, selection, threshold, branch or another resolution rule. Prior iteration and upstream state stay fixed. Fixed-map internal-coordinate evaluation alone is not RF-03. Equal resulting maps do not replace the declared causal query or its trace.

RF-04 covers exact fixed sequence/index; RF-05 includes stop/resource/selector dependence and comparison population; RF-06 includes named upstream reruns; RF-07 fixes upstream producer/state/generation and composes exact FRUIT-only extent with RF-03/04/05 references. A fixed derivative is not a substitute.

PRED-006/009/010 distinguish fixed-map internal coordinates, frozen components, re-resolved defining state, affine offsets and finite/rank/support-domain failure. Leaving the fixed domain produces unavailable response or an exact separately typed transition; it never silently acquires RF-03 identity. Unestablished source/state maps remain unavailable. Numerical comparisons are blocked; evidence artifacts are not produced.

Normative source: review-candidate core, PTC specialization; ASM-002–005; REQ-011–013; PRED-006/009/010.

6 Law-relative covariance, shift/bias and NOI evidence

EV-011. Register exact target-law identity, random variables, fixed objects, resolution/selection condition, omitted variation, support/domain and reference/gauge/null-space state before that law's random event. Use separate fixtures for physical/random-parent, explicitly fixed-realized-model and conditional NOI-randomization laws. The first retains random parent/model values; the second omits model learning and has zero model/cross blocks only within its narrower law; the third may fix realized parent/model/operator/support and exact source-law successful design resolution before the NOI member draw. Do not call them equivalent or reject all resolution conditioning indiscriminately.

FRUIT branch/completion/convergence/terminal-conditioned targets need separate method and estimand authority and remain blocked here. Check that generation labels do not hide selection or change the requested law/domain. Independently expand the exact complete-fixed-linear covariance under its finite-joint-moment premises; retain both cross terms unless exact law/conditioning authorizes omission. Transform accepted covariance through the declared application relation and dependence, keeping transformation uncertainty and all core-declared roles separate.

For PRED-005 compare the fixed-linear transfer operator D_k , model mismatch, applied-error-induced output shift and law-qualified bias as four distinct products. Bias needs exact estimand/reference/error-law/conditioning; no expectation authority means unavailable bias even if the transfer and shift are available. A deterministic discrepancy or shift is not covariance.

EV-012. Bind iteration/member/ensemble generation, exact target law, parent/output domains, center convention and exact external NOI-owned product, source and estimator identity/digest. Missing external bindings are explicitly unavailable and block numerical comparison. The owner correction supplies the conditional rule only. With fixed model/state and complete linearity, independently construct $A_k \delta y_{k,b}^{\text{NOI}}$ for a zero-target-law member and $A_k y_{k,b}^{\text{full}} + D_k m_k^{\text{applied}}$ for a full parent member with exact center. Do not add deterministic rejoin to the zero-target-law member.

A discriminator permits nonzero spatial/member sample mean and source/residual structure while its expectation under the declared law is zero. Require no finite-ensemble mean subtraction or $B - 1$ correction by implication. Full-affine center/estimator authority, source-law design-resolution conditioning and response/support parity are exact independent bindings. Preserve distinct fixed-state, dependent-successor and relearned methods/ensembles and member-state graphs; no pooling or route admission follows. Every numerical result here is **blocked**; evidence-artifact realization is separately recorded.

Normative source: review-candidate core, REQ-014–016; PRED-005, PRED-011, PRED-018.

7 Continuation, restart, retry and stopping evidence

EV-008. A prospective continuation fixture binds one exact approved method/profile, immutable base parent, future controls and every causal state query. Its independent expected relation compares uninterrupted execution with checkpoint/continuation after completed absolute zero-based iteration N . Continuation must start at $N + 1$ from S_{N+1} ; N is not a count. Compare all required scientific state/results, support/reference/gauge and stochastic state under the preregistered numerical equivalence policy, not only the final array. Record reconstruction and failure behavior for compact/lineage storage.

No method, stopping point, replay budget, numerical profile or tolerance is selected here. An actual replay requires separate authorized numerical work. A successful process exit is not evidence of continuation sufficiency. The fixture's completeness argument must identify all future-consequential facts, not merely the fields convenient to serialize.

EV-009. Prospective restart cases establish a new lineage and exact reused ancestry, initialization and counters. Relearning under unchanged rules produces new realized generations; a rule change creates a new method. Retry cases retain parent state, randomness/key/counters, attempt identity and earlier failure. Compare their independent identity graph and scientific state queries. Request-state fixtures distinguish immutable not-requested/requested generations and a later new request. Scientific availability and producer outcome are separate from requested-generation stage order. Early-stop fixtures inspect every reached stage and prove that no later realized/completed/published success was fabricated. A separate replacement may reference a failed attempt but cannot erase it.

EV-013. Inspect completion, convergence, resource limit, policy stop, failure, selection and fitness as separate records. A future convergence comparison binds its exact proposition, quantity/domain, metric/norm, scale/denominator, tolerance, direction, persistence/consecutive condition, missing/non-finite behavior, state and generation. Resource cases compare counters/accounting and exhaustion consequences independently. PRED-015's negative control reaches a limit without the convergence premise; the expected record reports limit/stop only.

These are prospective procedures and expected relations. No continuation, restart, retry, convergence or resource experiment has been run. Every numerical entry remains **blocked**, with its named method/profile, fixture and comparison dependencies explicit.

Normative source: review-candidate core, ASM-006; REQ-010, REQ-017–019; PRED-013–015.

8 Comparable selection, completion and immutable publication

EV-014. Bind an exact approved selector method, data access, candidate population and comparison policy before inspecting outcomes. Independently construct the common quantity/role, unit/scale, frame/grid/domain, reference/ gauge, response meaning, support comparison domain, method/generation relation and uncertainty/metric meaning. For cross-route/method/grouping/gauge/support candidates, retain every selector-owned conversion in its method identity. Equal-shape and common-scalar incompatible candidates are negative controls.

Compare every candidate's inclusion/exclusion, conversion, ties, unavailable comparison and cause. Apply only the exact predeclared rule; no latest, final, lowest-residual or visual preference is implied. Check all nine terminal fields, including selected completed identity, whole population, rule/generation, convergence/stop/resource facts, causes, response, uncertainty, continuation/ terminality and only explicitly requested/linked named-use identities/statuses or a no-request statement. Field 9 is open-world, not a list of every future use. Missing hard comparison/selection dependencies block the claim. Terminal evidence references the selected bundle without mutating it and retains RF-05 and any RF-07 composition scope.

EV-015. Compare the candidate's universal and method-required bundle roles against an independently constructed inventory and their permitted statuses. Check immutable request state and generation separately from the requested/effective/resolved/applied/realized/candidate/decision/completed/published producer path, scientific availability and branch/role outcomes. Verify exact core-declared product identities and bundle-bound scientific handoff: bare z_k requires its exact bundle or immutable reference. A complete candidate needs every required value or allowed typed status; only its exact FRUIT-owned completion rule establishes completed iteration. Publication exposes the immutable completed role and generation with provenance.

Use PRED-020 cases for open-world requested/linked child decisions or an explicit no-request record. Retain SCI-VAL request, applicability, eligibility and realization separately with exact profile/source. Not-requested creates no eligibility proposition. Missing profile/unresolved scope may produce a realized accounting record with applicability_unknown and decision_unavailable; incomplete/failed/not-produced artifacts are different realization conditions. Generic unavailable is not a SCI-VAL realization value. Check that base iteration/terminal completion survives missing or failed use and that only an explicitly requested narrower FRUIT role can require its specified child. Later children preserve immutable parent bytes/identity without a new iteration. SCI-VAL does not execute FRUIT or choose model/terminal.

Record structural evidence separately from numerical candidate conformance. This document creates no actual completed numerical iteration, selector, publication or downstream-admission result.

Normative source: review-candidate core, REQ-018-022; PRED-016, PRED-020.

9 Requirement-to-evidence index

Each row names an exact normative requirement and prospective evidence procedure(s) EV-001–015 above. Observable summaries are navigation, not replacement requirements. Every numerical result is **blocked**; the common blocking dependency is the absent separately approved exact numerical method and required route (in addition to unresolved fixture/oracle/comparison bindings).

REQ	EV	Observable / record discriminator
001	001	A/B/C decision binding
002	001, 004	Typed object/domain inventory
003	002	Three model stages and transformations
004	004	Removal/rejoin compatibility
005	003, 006	Complete path equation and counts
006	006, 010	Linearity and derivative premises
007	001	Route/grouping/input ancestry
008	005	Eight support/influence roles
009	005, 006	Occurrence and mode-origin propagation
010	007	Rules versus realized generations
011	003, 007	Defining-state and internal-coordinate role records
012	005, 006, 010	RF-01/02 fixed-map and affine PTC response
013	010	RF-03 defining-state reruns and RF-04–07 scopes
014	011	Law-relative target and four covariance blocks
015	002, 011	Core-declared uncertainty, transfer/shift/bias roles
016	012	NOI law, member meaning, centers and action
017	008, 009	Causal continuation/restart/retry state
018	009, 015	Request, availability, outcome and producer stage
019	013	Separate completion/stop/convergence
020	014	Comparable selector and open-world use references
021	006, 015	Bundle handoff and role-local dependencies
022	001, 005, 012, 015	Producer and SCI-VAL-congruent child records
023	001	Complete method record and approval gate
024	006, 010, 011, 012, 015	Complete r0.3 review-candidate core and claim ceilings

The full machine-readable index binds each complete ID and result separately. A future assessment records **pass**, **fail**, **blocked**, **not_applicable**, or **not_assessed** per requirement using the exact vocabulary spelling in the result contract; this table's current state is not a completed numerical conformance matrix. Applicability and every blocked dependency must remain explicit.

10 Prediction-to-fixture index

SCI-FRUIT-FIX-001–020 are prospective design identifiers, not existing numerical payloads. Each number maps one-to-one to the same PRED suffix. Independent expected constructions and blocked numerical results are in the delivered fixture index. No new scientific prediction ID is introduced.

ID	Prospective fixture discriminator
001	Zero applied operand; non-zero-preserving control
002	Identity residual and compatible equal removal/rejoin
003	Complete linearity with equal model-path maps
004	Perfect applied reference, distinct accepted-stage control
005	Distinct mismatch, transfer, shift and law-qualified bias
006	Fixed PTC map versus frozen component; affine centering
007	Permitted model-only and unavailable measured mode
008	Unequal support and incompatible reference/gauge
009	Internal fixed-map coordinates versus defining-state reruns
010	Finite/rank/support-domain failure without implicit RF-03
011	Physical-parent, fixed-model and conditional NOI laws
012	Repeated shared parent/model lineage
013	Checkpoint after absolute index N and same future controls
014	Immutable request generation and early branch/role outcomes
015	Declared resource exhaustion without convergence premise
016	Comparable selector and open-world linked use records
017	Observation/coadd/joint grouping comparison
018	Zero-target-law NOI members versus full-parent members
019	Missing/unapproved method or upstream authority
020	Bundle-bound payload and SCI-VAL-congruent child accounting

Each future fixture requires immutable payload/digest, exact applicable premises, independent expected construction and preregistered comparison. Null fixture payloads in this packet mean no numerical fixture has been instantiated or run. A symbolic deduction does not turn that unassessed numerical entry into pass.

11 Source, build and core parity procedure

Resolve the exact candidate source tree, build recipe/tool versions and runtime/environment identity. Rebuild from an isolated copy of the declared source closure, without inherited build intermediates. Bind produced artifacts and logs to the candidate tuple. Compare exact core identity and canonical source-inventory digest with the value authorized for that candidate. A digest mismatch creates a different or unresolved candidate; it is not silently waived.

For these documents, the canonical core digest is SHA-256 of the exact sorted source-inventory bytes listing six normative modules and the core body. Each view has a separate source-inventory digest covering its wrapper/body, common core modules and shared typography/cover. Generated digest bindings are derived from those inventories and excluded from their own scopes to avoid self-reference; the final artifact manifest binds their actual bytes. The identical core binding is imported into all three covers. Equations used in the rationale invoke the same canonical macros. The ECS cites their requirement/prediction identity.

The delivered build recipe records exact local Tectonic, cached TeX resources, Python/document packages and Poppler identities. Its clean build compiles an isolated declared source closure and compares output identities; the report states exactly whether bytes or a narrower content comparison agree. Reopen all PDFs, check metadata titles/owner/revision/date/status, resolve internal references, render all pages and inspect typography/equations/tables.

Verify gap-free REQ-001–024, ASM-001–006 and PRED-001–020; all product roles, lifecycle states, support roles, uncertainty distinctions and RF scopes; local link closure; all source and input hashes; exact archive membership and byte parity. The exact Stage A archive is a preserved opaque input, not permission to inspect manager-only science. The exact r0.2 delivery, including earlier historical inputs, remains preserved historical input, not relabeled as r0.3 output.

Document compilation, source parity, PDF rendering and a consistency review are document verification only. A missing source/digest blocks its dependent identity/conformance claim with scope and cause; it does not manufacture numerical authority or invalidate unrelated reproduced sources by implication.

Normative source: review-candidate core, REQ-001, REQ-010, REQ-023–024.

12 Blank result contract and limits of any future result

The delivered `evidence/CONFORMANCE_RESULT_TEMPLATE.json` supplies the following fields with unresolved candidate values and `blocked`:

Identity

record identity; evidence generation; requirement ID; candidate tuple and its digest/serialization convention; scientific core/method/owner bindings.

Observation

exact observable; quantity/units/domain/support/reference; actual artifact identities and digests.

Expected construction

immutable fixture identity/digest; independent oracle identity/digest, premises and dependence disclosure.

Comparison

preregistration identity/digest; comparison policy, numerical profile and required missing/non-finite or tolerance rules, left unresolved here.

Disposition

exact five-state result; separately typed evidence-artifact realization; applicability premise; blocking dependency/cause; deviations/failures; reviewer/date; affected claim/scope; superseded evidence parent.

The JSON is a blank evidence-record contract, not an implementation configuration or selected numerical schema. Nulls are unresolved. Replace them with immutable approved bindings only in a separately authorized assessment; retain old records when a new candidate or evidence generation supersedes them.

Keep these claims distinct in any result report:

Engineering conformance

Evidence for exact applicable core requirements and candidate, with every blocked/unassessed/failing item visible.

Response/representation fidelity

Agreement with its own exact scientific target and approved comparison; not implied by conformance.

Uncertainty coverage

A separate probability/population/coverage proposition and evidence; not inferred from a finite covariance.

Observational validation

Separately authorized observations and tests; no such activity is performed here.

Performance

Measured cost/throughput under an exact workload/environment, not a scientific correctness argument.

Readiness

Its own bounded operational decision; document completeness is not readiness.

Production authorization

Explicit authorized-use decision with its own owner/evidence; none is granted here.

The present numerical disposition is `blocked`. Evidence-artifact realization is separately typed; a realized accounting record is not numerical conformance. No route, implementation conformity, recovered measured mode, convergence adequacy, Pointing/OOF/catalogue fitness, production suitability/authorization or Unity activity is claimed. Independent consistency review remains advisory; Grant Wilson's possible conditional scientific freeze is a separate decision.